

Lista 7

1-a) $A = (5, 4, 1)$ e $B = (-2, 3, 2)$

$$U = (-2 - 5, 3 - 4, 2 - 1) \Rightarrow (-7, -1, 1)$$

$$P = A + tU \Rightarrow x = 5 + t \cdot (-7) \Rightarrow x = 5 - 7t$$

$$y = 4 + t \cdot (-1) \Rightarrow y = 4 - t$$

$$z = 1 + t \cdot 1 \Rightarrow z = 1 + t$$

forma Simetrica: $t = \frac{5-x}{-7}, t = 4-y, t = z-1$

$$\therefore z = \frac{5-x}{-7} = 4-y = z-1$$

b) $A = (0, -1, 0)$ e $B = (1, 0, 0)$

$$U = (1 - 0, 0 - (-1), 0 - 0) \Rightarrow (1, 1, 0)$$

$$P = A + tU \Rightarrow x = 0 + t$$

$$y = -1 + t$$

$$z = 0 + t \cdot 0 \Rightarrow z = 0$$

forma Simetrica: $t = x, t = y + 1, z = 0$

$$\therefore t = x = y + 1, z = 0$$

c) $A = (0, 1, -1)$ e $B = (0, 0, 0)$

$$U = (0 - 0, 0 - 1, 0 - (-1)) \Rightarrow (0, -1, 1)$$

$$P = A + tU \Rightarrow x = 0 + 0$$

$$y = 1 - t$$

$$z = -1 + t$$

forma Simetrica

$$x = 0, t = 1 - y, t = z + 1$$

$$\therefore t = 1 - y = z + 1, x = 0$$

$$d) A = (3, 2, 1) \quad , \quad B = (6, 1, -4)$$

$$\vec{v} = (6-3, 1-2, (-4)-1) = \vec{v} = (3, -1, -5)$$

$$P = A + t\vec{v} \Rightarrow x = 3 + 3t$$

$$y = 2 - 1t$$

$$z = 1 - 5t$$

forma simétrica

$$t = \frac{x-3}{3}, \quad t = \frac{y-2}{-1}, \quad t = \frac{z-1}{-5}$$

$$\therefore t = \frac{x-3}{3} = \frac{2-y}{-1} = \frac{1-z}{-5}$$

2-a) Para $\lambda = 0$

$$\begin{cases} x = 1 - 0 = 1 \\ y = 0 \\ z = 4 + 2 \cdot (0) = 4 \end{cases}$$

$$A = (1, 0, 4)$$

Para $\lambda = 1$

$$\begin{cases} x = 1 - 1 = 0 \\ y = 1 \\ z = 4 + 2 \cdot (1) = 6 \end{cases}$$

$$B = (0, 1, 6)$$

$$\vec{v}_1 = (-1, 1, 2)$$

$$\vec{v}_2 = (-2, 2, 4)$$

$$b) P = (1, 3, -3)$$

$$Q = (-3, 4, 12)$$

$$\begin{cases} x-1 = -\lambda \Rightarrow \lambda = 0 \\ y-3 = \lambda \Rightarrow \lambda = 3 \end{cases}$$

$$\begin{cases} -3 = 1 - \lambda \Rightarrow \lambda = 4 \\ 4 = 1 + \lambda \Rightarrow \lambda = 3 \\ 12 = -3 + 2\lambda \Rightarrow \lambda = 7.5 \end{cases}$$

P Não pertence a reta r

Q pertence a reta r

$$c) A = (1, 4, -7)$$

$$\vec{v} = (-1, 1, 2) \rightarrow \text{paralela r}$$

$$x = 1 + t \cdot (-1) \Rightarrow x = 1 - t$$

$$y = 4 + t \cdot 1 \Rightarrow y = 4 + t$$

$$z = -7 + t \cdot 2 \Rightarrow z = -7 + 2t$$

3- $A = (3, 6, -7)$, $B = (-5, 2, 3)$ e $C = (4, -7, -6)$

a) A, B, C Não colineares

$$AB = B - A = (-5 - 3, 2 - 6, 3 - (-7)) \Rightarrow (-8, -4, 10)$$

$$AC = C - A = (4 - 3, -7 - 6, -6 - (-7)) \Rightarrow (1, -13, 1)$$

$$AB = k \cdot AC \Rightarrow (-8, -4, 10) = k \cdot (1, -13, 1)$$

$$\begin{cases} -8 = k \cdot 1 \Rightarrow k = -8 \\ -4 = k \cdot (-13) \Rightarrow k = 4/13 \\ 10 = k \cdot 1 \Rightarrow k = 10 \end{cases}$$

$$\therefore -8 \neq 4/13 \neq 10$$

AB e AC Não paralelos

A, B e C Não são colineares

São Vértices de um triângulo

5)

$$M = \left(\frac{x_a + x_b}{2}, \frac{y_a + y_b}{2}, \frac{z_a + z_b}{2} \right) \Rightarrow \left(\frac{3 + (-5)}{2}, \frac{6 + 2}{2}, \frac{-7 + 3}{2} \right)$$

$$M = (-1, 4, -2)$$

$$CM = M - C = (-1 - 4, 4 - (-7), -2 - (-6)) \Rightarrow (-5, 11, 4)$$

$$r = C + t \cdot v$$

$$r = (4, -7, -6) + t(-5, 11, 4), \text{ onde } t \in \mathbb{R}$$

4-a) $C = (1 + \lambda, 2 + \lambda, -3\lambda)$

$$CA = A - C = (0 - (1 + \lambda), 1 - (2 + \lambda), 8 - (-3\lambda))$$

$$CA = (-1 - \lambda, -1 - \lambda, 8 + 3\lambda)$$

$$AB = B - A = (-3 - 0, 0 - 1, 9 - 8)$$

$$AB = (-3, -1, 1)$$

$$AB \cdot AC = 0 \Rightarrow (-3 \cdot (1 + \lambda) + (-1) \cdot (1 + \lambda) + 1 \cdot (-3\lambda - 8)) = 0$$

$$-4 - 4\lambda - 3\lambda - 8 = 0$$

$$-12 - 7\lambda \Rightarrow \lambda = -12/7$$

$$C = (1 + \lambda, 2 + \lambda, -3\lambda) = \left(1 - \frac{12}{7}, 2 - \frac{12}{7}, \frac{36}{7}\right)$$

$$C = \left(-\frac{5}{7}, \frac{2}{7}, \frac{36}{7}\right)$$

5) $\|X - A\| = \|X - B\| \quad X = (1 + \lambda, \lambda, \lambda)$

$$X - A = (1 + \lambda - 1, \lambda - 1, \lambda - 1) = (\lambda, \lambda - 1, \lambda - 1)$$

$$\|X - A\|^2 = \lambda^2 + (\lambda - 1)^2 + (\lambda - 1)^2 = \lambda^2 + 2(\lambda - 1)^2$$

$$X - B = (1 + \lambda - 0, \lambda - 0, \lambda - 1) = (1 + \lambda, \lambda, \lambda - 1)$$

$$\|X - B\|^2 = (1 + \lambda)^2 + \lambda^2 + (\lambda - 1)^2$$

$$\lambda^2 + 2(\lambda - 1)^2 = (1 + \lambda)^2 + (\lambda - 1)^2$$

$$\lambda^2 + \lambda^2 - 2\lambda + 1 = 1 + 2\lambda + \lambda^2$$

$$-4\lambda = 0$$

$$\lambda = 0$$

$$X = (1 + \lambda, \lambda, \lambda) \Rightarrow (1, 0, 0)$$

5-a) $\pi = X = A + \lambda \vec{v} + \mu \vec{w}$

$$\pi: (x, y, z) = (1, 2, 0) + \lambda(1, 1, 0) + \mu(2, 3, -1)$$

$$\begin{cases} x = 1 + \lambda + 2\mu, & \text{Equações Vetoriais:} \\ y = 2 + \lambda + 3\mu, \\ z = -\mu. \end{cases}$$

$$\pi = (x, y, z) = (1, 2, 0) + \lambda(1, 1, 0) + \mu(2, 3, -1)$$

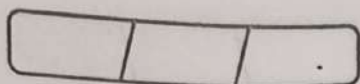
eq paramétrica

b) $\vec{AB} = B - A = (1 - 1, -1 - 1, -1 - 0) = (0, -2, -1)$

$$\pi = X = A + \lambda \vec{AB} + \mu \vec{v}$$

$$\begin{cases} x = 1 + 2\mu, \\ y = 1 - 2\lambda + \mu, \\ z = -\lambda \end{cases}$$

$$\pi = (x, y, z) = (1, 1, 0) + \lambda(0, -2, -1) + \mu(2, 1, 0)$$



$$c) \vec{AB} = B - A = (2-1, 1-0, -2-1) = (1, 1, -3)$$

$$\vec{AC} = C - A = (1-1, -1-0, 0-1) = (0, -1, -1)$$

$$\pi: x = A + \lambda \vec{AB} + \mu \vec{AC} \quad \vec{r} = (x, y, z) = (1, 0, 1) + \lambda(1, 1, -3) + \mu(0, -1, -1)$$

$$\begin{cases} x = 1 + \lambda, \\ y = -\mu, \\ z = 1 - 3\lambda - \mu. \end{cases}$$

6a)

$$\vec{n} = \vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = (1, 0, -1)$$

$$n_x(x-x_0) + n_y(y-y_0) + n_z(z-z_0) = 0$$

$$1(x-9) + 0(y+1) - 1(z-0) = 0$$

$$x - 9 - z = 0$$

$$\therefore \pi: x - 9 - z = 0$$

$$b) \vec{AB} = B - A = (-1-1, 0-0, 1-1) = (-2, 0, 0)$$

$$\vec{AC} = C - A = (2-1, 1-0, 2-1) = (1, 1, 1)$$

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = (0, 2, -2)$$

$$0 \cdot (x-1) + 2 \cdot (y-0) - 2 \cdot (z-1) = 0$$

$$2y - 2z + 2 = 0$$

$$y - z + 1 = 0$$

$$\therefore \pi: y - z + 1 = 0$$

$$c) \vec{AB} = B - A = (1-1, (-1)-1, (-1)-0) = (0, -2, -1)$$

$$n = \vec{AB} \times \vec{u} = \begin{vmatrix} i & j & k \\ 0 & -2 & -1 \\ 2 & 1 & 0 \end{vmatrix} = (1, -2, 4)$$

$$-2j \quad 4k \quad i$$

$$1(x-1) - 2(y-1) + 4(z-0) = 0$$

$$x-1 - 2y + 2 + 4z = 0$$

$$x - 2y + 4z + 1 = 0$$

$$\therefore \pi: x - 2y + 4z + 1 = 0$$

$$d) P = (1, -1, 1) \quad Q = (0, 2, 2)$$

$$PQ = Q - P = (0-1, 2+1, 2-1) = (-1, 3, 1)$$

$$n = \vec{PQ} \times \vec{v} = \begin{vmatrix} i & j & k \\ -1 & 3 & 1 \\ 1 & 1 & -1 \end{vmatrix} = (-4, 0, -4)$$

$$-4i - 4k - j$$

$$-4(x-1) + 0(y+1) - 4(z-1) = 0$$

$$-4x + 4 - 4z + 4 = 0$$

$$-4x - 4z + 8 = 0$$

$$x + z - 2 = 0$$

$$\therefore \pi: x + z - 2 = 0$$

$$7-0) y = \lambda \quad z = \mu$$

$$4x = -2y + z - 5$$

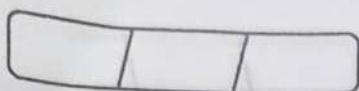
$$x = -\frac{1}{2}y + \frac{1}{4}z - \frac{5}{4}$$

$$x = -\frac{1}{2}\lambda + \frac{1}{4}\mu - \frac{5}{4}$$

$$\left. \begin{aligned} x &= -\frac{5}{4} - \frac{1}{2}\lambda + \frac{1}{4}\mu \\ y &= \lambda \\ z &= \mu \end{aligned} \right\}$$

$$y = \lambda$$

$$z = \mu$$



b) $5x - y - 1 = 0$

$$5x = y + 1$$

$$x = \frac{1}{5}y + \frac{1}{5}$$

$$x = \frac{1}{5} \cdot 1 + \frac{1}{5}$$

$$\begin{cases} x = \frac{1}{5} + \frac{1}{5} \lambda \\ y = 1, \\ z = \mu, \end{cases}$$

c) $z - 3 = 0$

$$z = 3$$

$$x = 1 \text{ e } y = \mu$$

$$\begin{cases} x = 1, \\ y = \mu, \\ z = 3. \end{cases}$$

d) $y - z - 2 = 0$

$$y = z + 2$$

$$x = 1 \text{ e } z = \mu$$

$$\begin{cases} x = 1, \\ y = 2 + \mu \\ z = \mu \end{cases}$$

8-a) $\begin{cases} x = 1 + 1 - \mu \\ y = 2\lambda + \mu \\ z = 3 - \mu \end{cases}$

$$A = (1, 0, 3)$$

$$\vec{v}^0 = (1, 2, 0) \text{ coef. } 1$$

$$\vec{u}^0 = (-1, 1, -1) \text{ coef. } \mu$$

$$\vec{n}^0 = \begin{vmatrix} i & j & k \\ 1 & 2 & 0 \\ 1 & 1 & -1 \end{vmatrix} = (-2, 1, 3)$$

$$-2(x-1) + 1(y-0) + 3(z-3) = 0$$

$$-2x + 2 + y + 3z - 9 = 0$$

$$-2x + y + 3z - 7 = 0$$

$$2x - y - 3z + 7 = 0$$

5) $\begin{cases} x = 1 + 1 \\ y = 2 \\ z = 3 - 1 + \mu \end{cases}$

$$A = (1, 2, 3)$$

$$\vec{v}^0 = (1, 0, -1) \text{ coef. } 1$$

$$\vec{u}^0 = (0, 0, 1) \text{ coef. } \mu$$

$$\vec{n}^0 = \begin{vmatrix} i & j & k \\ 1 & 0 & -1 \\ 0 & 0 & 1 \end{vmatrix} = (0, -1, 0)$$

$$0(x-1) - 1(y-2) + 0(z-3) = 0$$

$$-y + 2 = 0$$

$$y - 2 = 0$$

$$c) \begin{cases} x = -2 - \lambda - \mu & A = (-2, 0, 0) \\ y = 2\lambda + 2\mu & \vec{u} = (1, 2, 1) \text{ coef } \lambda \\ z = \lambda + \mu & \vec{v} = (-1, 2, 1) \text{ coef } \mu \end{cases}$$

$$n = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ -1 & 2 & 1 \end{vmatrix} = (0, -2, 4)$$

$$\begin{aligned} 0 \cdot (x+2) - 2(y-0) + 4(z-0) &= 0 \\ -2y + 4z &= 0 \\ y - 2z &= 0 \end{aligned}$$

$$9-a) \begin{cases} x = 1 + 2\lambda \\ y = \lambda \\ z = 1 + 3\lambda \end{cases}, S = \begin{cases} x = -1 + 4\mu \\ y = -1 + 2\mu \\ z = -2 + 6\mu \end{cases} = \begin{cases} 1 + 2\lambda = -1 + 4\mu \\ \lambda = -1 + 2\mu \\ 1 + 3\lambda = -2 + 6\mu \end{cases} \Rightarrow$$

$$\begin{aligned} 1 + 2(-1 + 2\mu) &= -1 + 4\mu & 1 + 3(-1 + 2\mu) &= -2 + 6\mu \\ 1 - 2 + 4\mu &= -1 + 4\mu & 1 - 3 + 6\mu &= -2 + 6\mu \\ -1 &= -1 & -2 &= -2 \end{aligned}$$

$$r(1, 0, 1) \text{ e } \lambda = 0 \Rightarrow \begin{cases} 1 = -1 + 4\mu \Rightarrow \mu = 0,5 \\ 0 = -1 + 2\mu \Rightarrow 1 = -1 + 2 \cdot 0,5 \Rightarrow 0 = 0 \\ 1 - 2 + 6\mu \Rightarrow 0 = -2 + 6 \cdot 0,5 \Rightarrow 1 = 1 \end{cases}$$

São Coincidentes, não há plano determinado

$$b) r: X = (1, 1, 0) + \lambda(1, 2, 3), S: X = (2, 3, 3) + \mu(3, 2, 1)$$

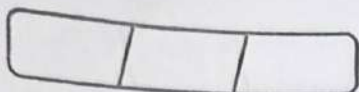
$$\begin{cases} 1 + \lambda = 2 + 3\mu \\ 1 + 2\lambda = 3 + 2\mu \\ 3\lambda = 3 + \mu \end{cases} \Rightarrow \begin{aligned} \mu &= 3\lambda - 3 \\ \mu &= 3 \cdot 1 - 3 \\ \mu &= 0 \end{aligned}$$

$$\begin{aligned} \text{II } 1 + \lambda &= 2 + 3 \cdot (3\lambda - 3) \\ 8 &= 8\lambda \\ \lambda &= 1 \\ \text{II } 3 \cdot 1 &\neq 3 + 0 \\ 3 &\neq 3, 0 \neq 0 \end{aligned}$$

$$P = (1 + 1, 1 + 2 \cdot 1, 0 + 3 \cdot 1) = (2, 3, 3)$$

(interseção $\lambda = 1$ em r)

$$\vec{u} = (1, 2, 3), \vec{v} = (3, 2, 1) \Rightarrow n = \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{vmatrix} = (-4, 8, -4)$$



$$-4(x-2) + 8(y-3) - 4(z-3) = 0$$

$$-4x + 8 + 8y - 24 - 4z + 12 = 0$$

$$-4x + 8y - 4z - 4 = 0$$

$$x - 2y + z + 1 = 0$$

concorrentes no ponto: $(2, 3, 3)$

Equação do plano: $x - 2y + z + 1 = 0$

c) $x = 2 - 4\lambda$

$y = 4 + 5\lambda$, $S: \frac{x}{2} = \frac{y-1}{-2} = z$

$z = 11$

$S \Rightarrow \frac{x}{2} = \mu \Rightarrow x = 2\mu$, $\frac{y-1}{-2} = \mu \Rightarrow y = 1 + 2\mu$, $z = \mu$

$S \begin{cases} x = 2\mu \\ y = 1 + 2\mu \\ z = \mu \end{cases} \quad S = r$
 $11 = \mu \Rightarrow x = 22, y = -21$

$22 = 2 - 4\lambda \Rightarrow \lambda = -5$

$-21 = 4 + 5 \cdot (-5) \Rightarrow -21 = -21$

interseção $(22, -21, 11)$

$r = \vec{u} = (-4, 5, 0)$

$S = \vec{v} = (2, -2, 1)$

$n = \begin{vmatrix} i & j & k \\ -4 & 5 & 0 \\ 2 & -2 & 1 \end{vmatrix} = (5, 4, -2)$

$n \cdot S(x-22) + 4(y+21) - 2(z-11) = 0$

$5x - 110 + 4y + 84 - 2z + 22 = 0$

$5x + 4y - 2z - 4 = 0$

concorrentes no ponto $(22, -21, 11)$

Equação do plano $5x + 4y - 2z - 4 = 0$

$$d) r: \frac{x-2}{3} = \frac{y+2}{4} = z \quad ; \quad s: \frac{x}{4} = \frac{y}{2} = \frac{z-3}{2}$$

$$\frac{x-2}{3} = \lambda \quad \frac{y+2}{4} = \lambda \quad z = \lambda \quad \left| \quad \frac{x}{4} = \mu, \frac{y}{2} = \mu, \frac{z-3}{2} = \mu \right.$$

$$x = 3\lambda + 2, \quad y = 4\lambda - 2, \quad z = \lambda \quad \left| \quad x = 4\mu, \quad y = 2\mu, \quad z = 2\mu + 3 \right.$$

$$r: \begin{cases} x = 3\lambda + 2 \\ y = 4\lambda - 2 \\ z = \lambda \end{cases} \quad s: \begin{cases} x = 4\mu \\ y = 2\mu \\ z = 3 + 2\mu \end{cases} \quad \Rightarrow \begin{cases} 3\lambda + 2 = 4\mu \\ 4\lambda - 2 = 2\mu \\ \lambda = 3 + 2\mu \end{cases}$$

$$I: 2 + 3(3 + 2\mu) = 4\mu \quad II: \lambda = 3 + 2(-5.5) \quad III: -2 + 4(-8) = 2(-5.5)$$

$$2 + 9 + 6\mu = 4\mu \quad \lambda = -8 \quad -2 - 32 = -11$$

$$2\mu = -11 \quad -34 = -11$$

$$\mu = -5.5 \quad \text{Falso}$$

$\therefore$ As retas não são concorrentes.

$$10-a) r: \begin{cases} x + 2y + 3z - 1 = 0 \\ x - y + 2z = 0 \end{cases} \quad I: (x + 2y + 3z - 1) - (x - y + 2z) = 0$$

$$3y + z - 1 = 0$$

$$z = 1 - 3y$$

$$II: x - y + 2(1 - 3y) = 0 \quad III: y = 1$$

$$x - y + 2 - 6y = 0$$

$$x - 7y + 2 = 0$$

$$x = 7y - 2$$

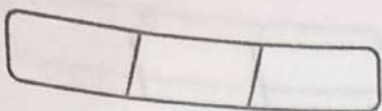
$$IV: x = 7\lambda - 2, y = 1, z = 1 - 3\lambda$$

$$V$$

$$\lambda = 0, \quad p = (-2, 0, 1)$$

$$v = (7, 1, -3)$$

$$\therefore r = x = (-2, 0, 1) + \lambda(7, 1, -3)$$



$$b) \begin{cases} x + y + z - 1 = 0 \\ x + y - z = 0 \end{cases} \Rightarrow (x + y + z - 1) - (x + y - z) = 0$$
$$2z - 1 = 0$$
$$z = \frac{1}{2}$$

$$x + y - \frac{1}{2} = 0 \quad y = 1$$

$$x = -y + \frac{1}{2}$$

$$x = \frac{1}{2} - 1, y = 1, z = \frac{1}{2}$$

$$P = \left(\frac{1}{2}, 0, \frac{1}{2}\right), \lambda = 0$$

$$V = (-1, 1, 0)$$

$$r = x - \left(\frac{1}{2}, 0, \frac{1}{2}\right) + \lambda \cdot (-1, 1, 0)$$

$$c) \begin{cases} x = 3 \end{cases}$$

$$2 \cdot (3) - z + 1 = 0$$

$$y = 1$$

$$r: \begin{cases} 2x - z + 1 = 0 \end{cases}$$

$$z = 7$$

$$x = 3, y = 1, z = 7$$

$$P = (3, 0, 7), \lambda = 0$$

$$V = (0, 1, 0)$$

$$r = x - (3, 0, 7) + \lambda (0, 1, 0)$$

$$d) \begin{cases} y = z \\ z = 0 \end{cases} \quad x = 1, y = 2, z = 0$$

$$P = (0, 2, 0), \lambda = 0$$

$$r = x - (0, 2, 0) + \lambda (1, 0, 0)$$

$$V = (1, 0, 0)$$

$$11-a) r: x = (1, -1, 1) + \lambda (-2, 1, -1), \quad S = \begin{cases} y + z = 3 \\ x + y - z = 6 \end{cases}$$

$$S: y = 3 - z, \text{ para } z = \mu$$

$$y = 3 - \mu$$

$$x = 6 - y + z$$

$$S: x = (3, 3, 0) + \mu (2, -1, 1)$$

$$x = 6 - (3 - \mu) + \mu$$

$$x = 3 + 2\mu$$

$$x = 3 + 2\mu, y = 3 - \mu, z = \mu$$

paralelas ou colineares:

$$\vec{v} = x \cdot \vec{u} \Rightarrow (-2, 1, -1) = x \cdot (2, -1, 1)$$
$$x = -1, \text{ ou seja } \vec{v} = -\vec{u}$$

São paralelas

Coincidentes? $r, p: (1, -1, 1)$

$$y + z = -1 + 1 = 0 \neq 3 \text{ Não pertence}$$

$\therefore$ retas paralelas distintas

b) $r: \frac{x+1}{2} = \frac{y}{3} = \frac{z+1}{2}, s: x = (0, 0, 0) + \lambda (1, 2, 0)$

$$x = 2\lambda - 1, y = 3\lambda, z = 2\lambda - 1$$

$$r: x = (-1, 0, -1) + \lambda (2, 3, 2)$$

$$\left. \begin{array}{l} -1 + 2\lambda = 1 \\ 3\lambda = 2\lambda \\ -1 + 2\lambda = 0 \end{array} \right\} \Rightarrow \lambda = 1, \lambda = 2, \lambda = 0.5 \neq \text{falso}$$

$$\vec{u} = x \vec{v} \Rightarrow (2, 3, 2) = x (1, 2, 0)$$

Não coincide

falso não paralelas

$\therefore$ As retas reversas

c) $r: x = (8, 1, 9) + \lambda (2, 1, 3), s: x = (3, -4, 4) + \mu (1, -2, 2)$

$$8 + 2\lambda = 3 + \mu \Rightarrow \mu = 5 + 2\lambda$$

$$1 - \lambda = -4 - 2\mu \Rightarrow 3\lambda = -15 \Rightarrow \lambda = -5$$

$$9 + 3\lambda = 4 + 2\mu$$

$$\mu = 5 + 2(-5)$$

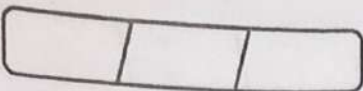
$$\mu = -5$$

$$9 + 3 \cdot (-5) = 4 + 2 \cdot (-5)$$

$$-6 = -6, \text{ verdadeiro}$$

$$P = (8 + 2(-5), 1 - (-5), 9 + 3(-5)) = (-2, 6, -6)$$

$\therefore$ As retas são concorrentes no ponto $(-2, 6, -6)$



$$d) r = \frac{x+1}{2} = y = z, \quad S: \begin{cases} x+y-3z=1 \\ 2x-y-2z=0 \end{cases}$$

$$x=2\lambda-1, y=\lambda, z=\lambda \quad \left| \quad \begin{aligned} 3x-5z &= 1 \\ x &= -1+5z \end{aligned} \right. \quad y = \frac{2+4z}{3}$$

$$r: (-1, 0, 0) + \lambda(2, 1, 1)$$

$$\text{Seja } z = 3\mu$$

Interseção:

$$S: X = \left(\frac{1}{3}, \frac{2}{3}, 0\right) + \mu(5, 4, 3)$$

$$\begin{cases} -1+2\lambda = \frac{1}{3} + 5\mu \\ \lambda = \frac{2}{3} + 4\mu \end{cases} \Rightarrow$$

$$\begin{aligned} 3\mu &= \frac{2}{3} + 4\mu & -1+2(3 \cdot \frac{2}{3}) &= \frac{1}{3} + 5(\frac{2}{3}) \\ \mu &= -\frac{2}{3} & -5 &= -3 \end{aligned}$$

falso

$$\text{Paralela: } \vec{v} = x \vec{u} \Rightarrow (2, 1, 1) = x \cdot (5, 4, 3)$$

Não interseção

Não são paralelas

∴ As retas são reversas

$$12-a) r = X = (1, 1, 0) + \lambda(0, 1, 1), \quad \pi: x-y-z=2$$

$$\vec{v} = (0, 1, 1)$$

$$\vec{n} = (1, -1, -1) \quad \vec{v} \cdot \vec{n} = 0 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) = -2 \neq 0 \text{ transversais em } \pi$$

$$X = 1 + 0\lambda = 1; y = 1 + \lambda, z = 0 \Rightarrow 1 - (1 + \lambda) - 0 = 2$$

$$-2\lambda = 2 \Rightarrow \lambda = -1$$

$$\text{para } \lambda = -1, P = (1, 1 + (-1), -1) = (1, 0, -1)$$

∴ Retas são transversais, com interseção em $(1, 0, -1)$

$$5) \frac{x-1}{2} = y = z, \quad \pi: X = (3, 0, 1) + \lambda(1, 0, 1) + \mu(2, 2, 0)$$

$$x = 2\lambda + 1, y = \lambda, z = \lambda$$

$$\pi: X = (1, 0, 0) + \lambda(2, 1, 1)$$

$$\vec{v} = (2, 1, 1)$$

$$n = \begin{vmatrix} i & j & k \\ 1 & 0 & 1 \\ 2 & 0 & 2 \end{vmatrix} = (-2, 2, 2)$$

$$\vec{v} \cdot \vec{n} = 2 \cdot (-2) + 1 \cdot 2 + 1 \cdot 2 = 0 \text{ paralela}$$

$$r = (1, 0, 0)$$

$$1 = 3 + 1 + 2\mu, \quad 0 = 0 + 2\mu, \quad 0 = 1 + 1 + 0$$

$$1 = 3 + 1 + 2 \cdot 0$$

$$\mu = 2$$

$$1 = -1$$

$$1 = 2$$

falso

$\therefore$ Reta é paralela ao plano e não está contida nele

$$c) r: \begin{cases} x - y = 1 \\ x - 2y = 0 \end{cases}, \quad \pi: x + y = 2$$

$$n = (1, 1, 0)$$

$$0 \cdot y = 1$$

$$x - 1 = 1 \Rightarrow x = 2$$

$$\vec{v} \cdot \vec{n} = 0 \cdot 1 + 0 \cdot 1 + 1 \cdot 0 = 0$$

paralela

$$r: (2, 1, 0) + \lambda(0, 0, 1)$$

$$P: (2, 1, 0) \Rightarrow x + y = 2 \Rightarrow 2 + 1 = 2$$

$$3 = 2 \text{ falso}$$

$\therefore$ A reta é paralela ao plano e não está contida nele

$$d) r: x - 2y = 3 - 2z + y \Leftrightarrow 2x - z = 3, \quad \pi: x = (1, 4, 0) + \lambda(1, 1, 1) + \mu(2, 1, 0)$$

$$\begin{cases} x - 3y + 2z = 3 \\ -2z + y - z = -3 \end{cases}$$

$$n = \begin{vmatrix} i & j & k \\ 1 & -3 & 2 \\ 0 & 1 & -1 \end{vmatrix} = (-1, 2, -1)$$

$$y = -3x + 3$$

$$x - 3(-3x + 3) + 2z = 3$$

$$10x + 2z = 12$$

$$z = 6 - 5x$$

$$n \cdot \vec{v} = 1 \cdot (-1) + (-3) \cdot 2 + (-5) \cdot (-1) = -2 \neq 0$$

transversal

$$\text{seja } x = \lambda$$

$$P = (1, 4, 0) \quad n = (-1, 2, -1)$$

$$x = \lambda, \quad y = -3\lambda + 3, \quad z = 6 - 5\lambda$$

$$-1 \cdot (x - 1) + 2(y - 4) - 1 \cdot (z - 0) = 0$$

$$r = x(0, 3, 6) + \lambda(1, -3, -5)$$

$$-x + 1 + 2y - 8 - z = 0$$

$$-x + 2y - z - 7 = 0$$

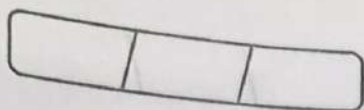
$$x - 2y + z + 7 = 0$$

$$-\lambda + 2(-3\lambda + 3) - (6 - 5\lambda) - 7 = 0$$

$$-2\lambda - 7 = 0 \Rightarrow \lambda = -7/2$$

$$P = \left(-\frac{7}{2}, -3 - \left(-\frac{7}{2}\right) + 3, 6 - 5\left(-\frac{7}{2}\right)\right) \Rightarrow \left(-\frac{7}{2}, \frac{27}{2}, \frac{47}{2}\right)$$

$\therefore$ Reta e o plano são transversais, com interseção em $\left(-\frac{7}{2}, \frac{27}{2}, \frac{47}{2}\right)$



13-a) $r: x = (1, 1, 1) + \lambda(2, m, 1)$ e $\pi: x(0, 0, 0) + \alpha(1, 2, 0) + \beta(1, 0, 1)$

$$\vec{n} = \vec{u} \times \vec{v} = \begin{vmatrix} i & j & k \\ 1 & 2 & 0 \\ 1 & 0 & 1 \end{vmatrix} = (2, -1, -2)$$

paralelas: $\vec{v} \cdot \vec{n} = 0$

$$(2, m, 1) \cdot (2, -1, -2) = 0 \Rightarrow 2 \cdot 2 - m - 2 = 0 \Rightarrow m = 2$$

b) $r: x(n, 2, 0) + \lambda(2, m, m)$, $\pi: x - 3y + z = 1$

$$\vec{v} = (2, m, m)$$

$$\vec{n} = (1, -3, 1)$$

$$\vec{v} \cdot \vec{n} = 2 \cdot 1 - 3m + m = 0$$

$$-2m = -2$$

$$m = 1$$

$$(n, 2, 0) \rightarrow x - 3y + z = 1$$

$$n - 3 \cdot 2 + 0 = 1$$

$$n = 7$$

c) $\frac{x}{m} - \frac{y}{2} = \frac{z}{m}$, $\pi: x + my + z = 0$

$$\vec{n} = (1, m, 1)$$

$$x = m\lambda + 1, y = 2\lambda, z = m\lambda$$

$$r: x(1, 0, 0) + \lambda(m, 2, m)$$

$$\vec{v} \cdot \vec{n} = 0$$

$$m + 2m + m = 0$$

$$4m = 0$$

$$m = 0$$

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$$\pi_1: x = (4, 2, 4) + \lambda(1, 1, 2) + \mu(3, 3, 1) \quad \pi_2: x = (3, 0, 0) + \lambda(1, 1, 0) + \mu(0, 1, 4)$$

$$\vec{n}_1 = \begin{vmatrix} i & j & k \\ 1 & 1 & 2 \\ 3 & 3 & 1 \end{vmatrix} = (-5, 5, 0)$$

$$\vec{n}_2 = \begin{vmatrix} i & j & k \\ 1 & 1 & 0 \\ 0 & 1 & 4 \end{vmatrix} = (4, -4, 1)$$

$$\vec{n}_1 \cdot \vec{n}_2 = (-5, 5, 0) \cdot (4, -4, 1) = -20 + 20 + 0 = 0 \quad \text{Não } \Rightarrow \text{transversais}$$

$$-5(x-4) + 5(y-2) + 0(z-4) = 0$$

$$4(x-3) - 4(y-0) + 1(z-0) = 0$$

$$x - y = 2$$

$$4x - 4y + z - 12 = 0$$

$$\begin{cases} x - y = 2 \\ 9x - 4y + z = 12 \\ z = 4 \end{cases} \quad \text{Retas transversais:} \quad x = (2, 0, 4) + \lambda(1, 1, 0)$$

b) $\pi_1: x - y + 2z - 2 = 0$, $\pi_2: x = (0, 0, 1) + \lambda(1, 0, 3) + \mu(-1, 1, 1)$
 $n_1 = (1, -1, 2)$, $n_2 = \begin{vmatrix} 1 & 0 & 3 \\ 1 & 0 & 3 \\ -1 & 1 & 1 \end{vmatrix} = (-3, -4, 1)$
 n_1 e $n_2 \rightarrow$ Não são múltiplas
transversais

$$\begin{cases} x - y + 2z = 2 & \rightarrow x = 2 + y - 2z & -3(x - 0) - 4(y - 0) + 1(z - 1) = 0 \\ -3x - 4y + z = 1 & \rightarrow -3(2 + y - 2z) - 4y + z = 1 \end{cases}$$

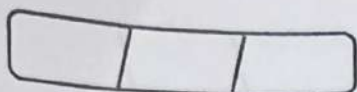
$$x = (1, 0, 0, 5) + \lambda(9, 7, 1)$$

c) $\pi_1: 2x - y + z - 1 = 0$, $\pi_2: 4x - 2y + 2z - 9 = 0$
 $n_1(2, -1, 1)$, $n_2(4, -2, 2)$ $-1 \neq -9$
 $n_1 = \frac{1}{2} \cdot n_2$ plano paralelos Distintos

d) $\pi_1: A(0, 1, 6), B(5, 0, 1) \text{ e } C(4, 0, 0)$ $\pi_2: 4x + 40y - 4z - 16 = 0$
 $\vec{AB} = B - A = (5 - 0, 0 - 1, 1 - 6) = (5, -1, -5)$ $n_2 = (4, 40, -4) \rightarrow$
 $\vec{AC} = C - A = (4 - 0, 0 - 1, 0 - 6) = (4, -1, -6)$ $n_2 = (1, 10, -1) \rightarrow$

$$n_1 = \begin{vmatrix} i & j & k \\ 5 & -1 & -5 \\ 4 & -1 & -6 \end{vmatrix} = (1, 10, -1) \quad 1 \cdot n_1 = n_2 \Rightarrow \text{paralelos coincidentes}$$

15-a) $\pi_1 \begin{cases} x = -\lambda_1 + 2\mu_1 \\ y = m\lambda_1 \\ z = \lambda_1 + \mu_1 \end{cases}$ $\pi_2 \begin{cases} x = 1 + m\lambda_2 + \mu_2 \\ y = 2\lambda_2 \\ z = 3 + m\mu_2 \end{cases}$
 $\vec{v}_1 = (-1, m, 1)$ $\vec{v}_2 = (m, 2, 0)$
 $\vec{\mu}_1 = (2, 0, 1)$ $\vec{\mu}_2 = (1, 0, m)$



$$n_1: v_1 \times u_1 \begin{vmatrix} i & j & k \\ -1 & m & 1 \\ 2 & 0 & 1 \end{vmatrix} = (m, 3, -2m)$$

$$n_2: v_2 \times u_2 \begin{vmatrix} i & j & k \\ m & 1 & 0 \\ 1 & 0 & m \end{vmatrix} = (m, -m^2, -1)$$

$$(m, 3, -2m) = K(m, -m^2, -1)$$

$$\begin{cases} m = Km \\ 3 = Km^2 \\ -2m = -K \end{cases} \Rightarrow m \neq 0, K = 1$$

$$\Rightarrow 3 = 1 \cdot \left(\frac{1}{2}\right)^2$$

$$3 = -\frac{1}{4} \text{ (falso)}$$

$$m = \frac{1}{2}$$

Se $m = 0$, então $n_1(0, 3, 0)$ e $n_2(0, 0, -1)$
Não são paralelos.

$\therefore$ Para qualquer m real os planos π_1 e π_2 não são transversais.

b) $\pi_1 = (1, 1, 0) + \lambda(m, 1, 1) + \mu(1, 1, m)$ e $\pi_2 = 2x + 3y + 2z + n = 0$

$$n_1 \begin{vmatrix} i & j & k \\ m & 1 & 1 \\ 1 & 1 & m \end{vmatrix} = (m-1, -m^2+1, m-1)$$

$$n_2 = (2, 3, 2)$$

$$n_1 = K \cdot n_2 \Rightarrow (m-1, -m^2+1, m-1) = K \cdot (2, 3, 2)$$

$$\begin{cases} m-1 = 2K \\ -m^2+1 = 3K \\ m-1 = 2K \end{cases} \Rightarrow \begin{cases} m-1 = 2K \\ -m^2+1 = 3 \cdot \frac{m-1}{2} \end{cases} \Rightarrow \Delta = 9 - 40 = -31$$

$$2m^2 + 3m - 5 = 0$$

$$m = \frac{-3 \pm 7}{4} \Rightarrow m = \frac{4}{4} = 1 \text{ ou } m = \frac{-10}{4} = -\frac{5}{2}$$

Distintos: para $m = 1$

$n_1 = (0, 0, 0) \rightarrow$ Não é plano válido

$$\therefore m = -\frac{5}{2}, n \neq -5$$

Para $m = -\frac{5}{2}$ $p = (1, 1, 0)$

FORONI $2 \cdot 1 + 3 \cdot 1 + 2 \cdot 0 + n \neq 0 \Rightarrow n \neq -5$